

Stable Growth Consumer Staples ETF

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Investment Philosophy

This project aims to develop an actively managed ETF, called the Stable Growth Consumer Staples ETF (SGCS), that targets steady capital appreciation with lower downside risk by actively managing allocations between large-cap U.S. consumer staples stocks.

The SGCS ETF appeals to investors who want stability, income, and capital preservation rather than aggressive growth. This ETF would be an excellent option for large institutional investors seeking steadier returns such as pensions or endowments as a core defensive position. Likewise, this ETF is similarly sufficient option for conservative individual investors such as retirees or near-retirees that are looking for funds that preserve wealth during market downturns.

There are several companies that offer consumer staples ETFs [1] [2] [3]:

State Street	Consumer Staples Select Sector SPDR Fund (XLP)	~16 billion AUM
Vanguard	Vanguard Consumer Staples ETF (VDC)	~9 billion AUM
BlackRock	iShares U.S. Consumer Staples ETF (IYK)	~1 billion AUM

Each of these ETFs have a similar focus on large-cap U.S. consumer staples equities, however, they are all passively managed with quarterly rebalancing. This leaves them vulnerable to quickly evolving market conditions that may significantly decrease the value of the underlying equities.

A key component of the SGCS investment philosophy is its risk management discipline, which prioritizes capital preservation alongside steady compounding. The portfolio is comprised of five large-cap consumer-staples: Procter & Gamble, Coca-Cola, PepsiCo, Walmart and Costco. The long-term performance objective is to outperform the Vanguard Consumer Staples ETF (\$VDC) by approximately 3-5 percentage points in annualized compound annual growth rate (CAGR) while maintaining comparable drawdowns. In this way, the fund aims to achieve superior compounding without sacrificing the low-risk, defensive profile expected from a consumer-staples strategy.

Investment Methods

The SGCS ETF investment process begins with a curated universe of large-cap U.S. consumer staples equities obtained from Yahoo! Finance [4] [12]. These securities were selected for their high liquidity, societal relevance, and consistent earnings profiles. The underlying philosophy is to let data drive rebalancing decisions ensuring repeatability, transparency, and resistance to behavioral bias.

Modeling Framework

The core of the SGCS ETF centers on two quantitative strategies: momentum and mean reversion [5] [6]. Each approach reflects a distinct market inefficiency.

Momentum assumes that equities exhibiting strong recent performance will continue to outperform in the near term. The algorithm ranks securities based on trailing return strength and takes long positions in those within the top percentile.

Key hyperparameters include:

- Lookback period (window size for momentum calculation)
- Long/short designation (restricts strategy to long-only exposure)
- Neutral band (threshold defining when the market is trendless)

Mean Reversion assumes that asset prices revolve around a fair-value price. When prices fall below their moving average, the model signals a buy. When they exceed the upper quantile of recent prices, it signals a sell or reduction in position size.

Hyperparameters include:

- Lookback period (moving-average horizon)
- Quantile threshold (percentage of assets traded per period)
- Long/short designation

Each strategy's parameters were optimized through a grid search during the 2000–2019 training period, using Sharpe ratio as the primary performance objective. The optimization

routine evaluated hundreds of parameter combinations to identify configurations yielding the best risk-adjusted returns. Once optimal parameters were identified, they were applied without modification to the 2020–2024 testing period to assess true performance.

Trading Rules and Execution

The ETF follows a daily signal generation process that updates each equity's target weight based on the chosen model's signal (momentum or mean reversion). Trades are executed at the next available market close assuming perfect liquidity. To reduce transaction costs and prevent excessive turnover, a minimum signal strength filter ensures that only statistically significant deviations trigger trades. Transaction costs were modeled as 0 bps in the base case but can be adjusted for sensitivity analysis.

Exposure constraints are defined to maintain conservative positioning. The ETF operates as a long-only portfolio, eliminating short exposure after Monte Carlo testing revealed that shorting offered negligible upside with greater downside volatility. Portfolio weights are re-normalized at each rebalance to maintain full investment of available capital.

Performance Measurement

Performance metrics such as CAGR, Sharpe ratio, volatility, maximum drawdown, and alpha/beta relative to \$VDC are computed on both historical back tests and walk-forward simulations. This ensures that all trading rules are evaluated not only for profitability but also for stability, persistence, and robustness across different market periods. These methods form the foundation of a robust active management process.

Securities

SGCS is composed of a small basket of large-cap U.S. consumer-staples equities with consistent earnings, strong dividend histories, and resilient market share. The initial fund composition includes Procter & Gamble (\$PG) [7], Coca-Cola (\$KO) [8], PepsiCo (\$PEP) [9], Costco (\$COST) [10], and Walmart (\$WMT) [11]. These companies represent the most recognizable household goods, food, beverage, and retail sectors of the consumer-staples industry. These companies were chosen for their liquidity and role as sector leaders ensuring that the ETF maintains both defensive and investable characteristics suitable for institutional and individual investors alike.

Equity	Avg. Volume	Market Cap	PE Ratio
\$PG	7M	350B	21
\$KO	16M	300B	23
\$PEP	7M	200B	27
\$COST	2M	400B	50
\$WMT	16M	820B	39

Note: Metrics as of November 11, 2025

Performance Evaluation

The performance evaluation of the SGCS ETF examines both its historical performance relative to \$VDC and its forward-looking risk characteristics under simulated conditions. Through a combination of traditional back testing, walk-forward validation, and Monte Carlo analysis, the fund demonstrates consistent alpha generation, stable compounding, and resilience across market cycles.

The historical back test [13] (2000–2019 train against 2020–2024 test) reveals that the mean-reversion strategy with a 10-day lookback period, 40% quantile threshold, and long-only exposure produced the strongest results among all model configurations.

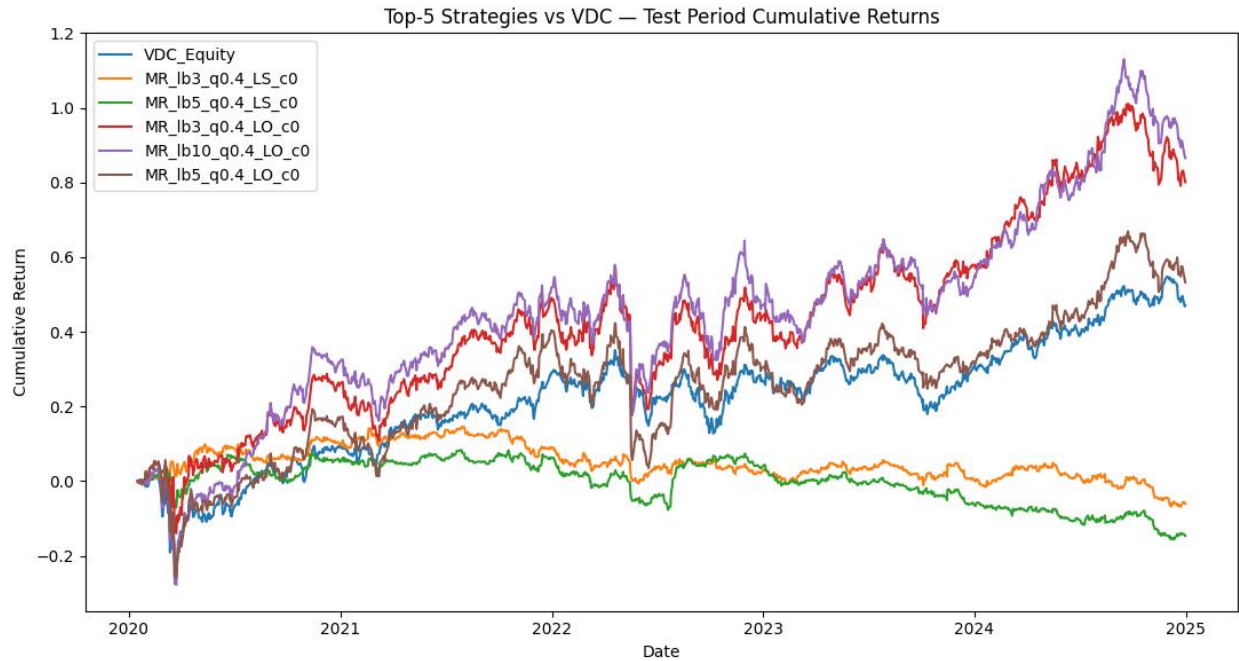


Figure 1: Top 5 Performing Strategies Tuned From 2000-2019 Against 2020-2024 vs \$VDC

This strategy achieved a CAGR of approximately 13%, a Sharpe ratio of 0.70, and volatility of 21%, compared to \$VDC's CAGR of 8%, Sharpe ratio of 0.54, and volatility of 17%. The strategy's maximum drawdown was around 30%, modestly higher than \$VDC's 25%, but it consistently recovered faster and sustained higher terminal wealth. These results demonstrate that the SGCS model delivered superior returns without a proportionate increase in risk, resulting in meaningfully higher risk-adjusted performance.

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=== Test Summary (Top-5 vs VDC) ===
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Series	CAGR	Sharpe	Vol	MaxDD	EndValue
---	---	---	---	---	---
str	f64	f64	f64	f64	f64
MR_lb10_q0.4_LO_c0	0.134146	0.703087	0.211161	-0.306924	1.865273
MR_lb3_q0.4_LO_c0	0.126083	0.68183	0.205255	-0.242262	1.800515
VDC_Equity	0.081073	0.544875	0.169652	-0.253069	1.471167
MR_lb5_q0.4_LO_c0	0.089991	0.524529	0.204691	-0.297765	1.53226
MR_lb3_q0.4_LS_c0	-0.012576	-0.117612	0.080305	-0.187459	0.939246
MR_lb5_q0.4_LS_c0	-0.031511	-0.338234	0.08425	-0.221109	0.853366

Figure 2: Summary Statistics of Top 5 Models vs \$VDC

The walk-forward back testing provided a deeper validation of the strategy's robustness over time. Using a rolling 5-year training window followed by a 1-year out-of-sample test, the model recalibrated annually and produced a consistent pattern of strong results across multiple market periods. The walk-forward analysis showed that the SGCS ETF maintained a positive CAGR and Sharpe ratio in most test years, with minimal deterioration in performance despite changing market conditions.

	TrainStartYear	TrainEndYear	TestYear	Strategy	MR_lookback	MR_quantile	MR_Jung_short	MOM_lookback	MOM_Jung_short	MOM_neutral_bond	trans_cost_bps	Test_CAGR	Test_Sharpe	Test_Volatility	Test_Max Drawdown	Test_Win Rate	Test_Avg Daily Turnover
0	2000	2004	2005	MR	3	0.400000	True	None	None	None	0.000000	3.95%	0.74	5.46%	-2.92%	52.21%	38.44%
1	2001	2005	2006	MR	5	0.400000	True	None	None	None	0.000000	2.46%	0.46	5.59%	-3.71%	48.37%	29.39%
2	2002	2006	2007	MR	10	0.400000	True	None	None	None	0.000000	12.02%	2.01	5.72%	-2.69%	55.42%	22.04%
3	2003	2007	2008	MR	10	0.400000	False	None	None	None	0.000000	-3.15%	0.02	27.88%	-20.05%	55.37%	24.76%
4	2004	2008	2009	MR	5	0.400000	True	None	None	None	0.000000	11.38%	1.27	8.79%	-5.18%	52.85%	30.07%
5	2005	2009	2010	MR	5	0.200000	True	None	None	None	0.000000	15.63%	1.94	7.65%	-5.75%	52.44%	42.45%
6	2006	2010	2011	MR	5	0.200000	True	None	None	None	0.000000	-4.42%	-0.45	9.05%	-8.26%	49.80%	39.43%
7	2007	2011	2012	MR	5	0.200000	True	None	None	None	0.000000	-11.88%	-1.41	8.69%	-14.53%	42.62%	34.36%
8	2008	2012	2013	MR	5	0.200000	True	None	None	None	0.000000	11.82%	1.48	7.74%	-5.25%	56.10%	44.49%
9	2009	2013	2014	MR	10	0.200000	False	None	None	None	0.000000	27.00%	2.15	11.44%	-7.21%	66.39%	32.08%
10	2010	2014	2015	MR	10	0.200000	False	None	None	None	0.000000	-5.53%	-0.33	14.12%	-16.60%	53.11%	30.83%
11	2011	2015	2016	MR	5	0.400000	False	None	None	None	0.000000	19.73%	1.76	10.55%	-4.04%	67.61%	32.52%
12	2012	2016	2017	MR	5	0.400000	False	None	None	None	0.000000	13.58%	1.49	8.78%	-8.36%	66.26%	30.61%
13	2013	2017	2018	MR	3	0.200000	False	None	None	None	0.000000	7.77%	0.50	18.04%	-12.45%	58.70%	48.78%
14	2014	2018	2019	MR	10	0.400000	False	None	None	None	0.000000	35.67%	3.31	9.35%	-5.64%	70.12%	26.39%
15	2015	2019	2020	MR	3	0.400000	False	None	None	None	0.000000	28.09%	1.06	26.58%	-14.68%	62.65%	38.84%
16	2016	2020	2021	MR	3	0.400000	False	None	None	None	0.000000	13.22%	1.29	9.99%	-7.32%	68.15%	38.33%
17	2017	2021	2022	MR	3	0.400000	False	None	None	None	0.000000	16.57%	1.10	14.94%	-8.78%	62.10%	37.79%
18	2018	2022	2023	MR	3	0.400000	False	None	None	None	0.000000	7.93%	0.79	10.33%	-10.84%	64.78%	37.26%
19	2019	2023	2024	MR	3	0.400000	False	None	None	None	0.000000	15.92%	1.67	9.09%	-7.14%	59.68%	38.87%

Figure 3: Summary Statistics of Walk-Forward Back Testing

The aggregated walk-forward equity curve confirmed that the strategy outperformed the benchmark steadily rather than relying on isolated periods of luck. Furthermore, the regression of strategy returns on benchmark returns yielded an annualized alpha of approximately 9% and a beta of 0.7, confirming that the SGCS portfolio generated excess returns independent of benchmark exposure while maintaining lower systematic risk.



Figure 4: Returns of SGCS ETF through Walk Forward Back Testing

To assess potential future outcomes, the top-performing mean-reversion model was passed through a Monte Carlo simulation using historical mean and volatility estimates from the test period. The simulation generated 500 random 5-year paths under three rebalancing frequencies - daily, weekly, and monthly - and compared them with equivalent \$VDC simulations. Across all frequencies, the SGCS ETF achieved a higher mean terminal value. The average terminal multiple was approximately 110% of the starting capital for SGCS versus 63% for \$VDC. Although the model exhibited slightly greater short-term volatility, its risk-return tradeoff remained superior due to its higher expected return and lower beta.

	MeanTerminal	StdTerminal	MeanReturn(%)	Volatility(Daily)	AvgMaxDD(%)	VaR_5%	CVaR_5%
Rebalance							
Benchmark (VDC)	\$1,630,193	\$642,968	63.02	0.0107	-27.94	\$848,987	\$741,001
Daily (Strategy)	\$2,151,771	\$1,153,851	115.18	0.0134	-31.49	\$841,134	\$680,965
Weekly (Strategy)	\$2,108,856	\$1,021,805	110.89	0.0134	-31.60	\$882,472	\$728,818
Monthly (Strategy)	\$2,060,719	\$1,078,864	106.07	0.0134	-31.83	\$875,414	\$699,553

Figure 5: Returns of Top Strategy with Varying Trade Frequencies vs \$VDC

Together, these evaluations demonstrate that the SGCS ETF effectively balances defensive investing with active management. The model has proven its ability to outperform the leading consumer-staples benchmark across multiple timeframes and market periods. The consistent positive alpha, moderate beta, and favorable simulated forward performance support

the conclusion that SGCS represents a quantitatively sound and operationally viable active ETF strategy capable of delivering stable growth for investors.

Management Recommendation

Based on the analytical evidence gathered through back testing, walk-forward validation, and Monte Carlo simulations, the SGCS ETF shows potential as a viable investment product. The strategy's consistent outperformance relative to \$VDC supports the case for developing this actively managed fund. From a management perspective, I would recommend the creation of the SGCS ETF. I would take the role of research lead leveraging my background in data science and analytics to oversee model development and strategy.

I believe this fund would provide a great investment vehicle for defensive portfolios for retirees, pensions, or endowments, but as an investor on the younger side, I would personally prefer to invest in ETFs with more growth potential.

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